From DNA to Protein

Amino acids, proteins, DNA, RNA, the central dogma, transcription and translation

1. Proteins

Proteins are the workers of the cell. Scientists estimate that at least 10,000 different proteins make you who you are and keep you that way (estimates vary depending on how they are counted). They are found in muscle, bone, skin, hair and almost every other tissue.

Job	Example protein	What it does
Enzymes	Amylase, pepsin	Speed up chemical reactions (digestion, copying DNA…)
Transport	Haemoglobin	Carries oxygen in the blood
Structure	Collagen, keratin	Gives strength to skin, bones, hair and nails
Defence	Antibodies	Recognise and neutralise germs
Messengers	Insulin	Carries signals between organs
Movement	Actin, myosin	Make muscles contract

Four levels of protein structure

- **Primary** — the sequence of amino acids (like letters in a word).
- **Secondary** — local folds such as the α-helix and β-sheet.
- **Tertiary** — the full 3D shape of one chain. Shape decides function.
- **Quaternary** — several chains working together (haemoglobin has four).

TRY IT: HOW MUCH PROTEIN DO YOU NEED?

Adults need at least 0.8 g of protein per kilogram of body weight per day (National Academy of Medicine). Growing teenagers need somewhat more.

Example: a 50 kg adult needs 0.8 × 50 = 40 g per day. Now calculate your own.

2. Amino acids

Amino acids are the building blocks of proteins. About 300 different amino acids exist in nature, but only 20 are used to build proteins in living things.

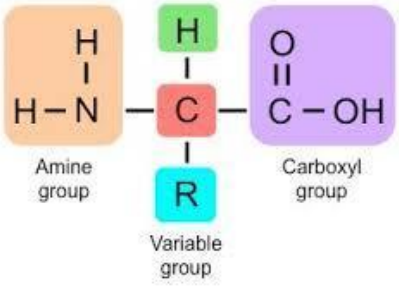


Figure 1. General structure of an amino acid. The central carbon (the α-carbon) holds four groups: an amino group, a carboxyl group, a hydrogen atom and a variable R-group. *Source: BioNinja*

Every amino acid has the same amino group ($-NH_2$), carboxyl group ($-COOH$) and hydrogen. Only the R-group (side chain) is different, and it gives each amino acid its personality: some are electrically charged, some repel water, some are bulky. The simplest amino acid, glycine, has just a hydrogen atom as its R-group.

The peptide bond

Amino acids join by a peptide bond: the carboxyl group of one reacts with the amino group of the next, and a molecule of water is released. A chain of amino acids is a polypeptide. One end of the chain has a free amino group (the N-terminus) and the other a free carboxyl group (the C-terminus). Proteins are always built from the N-terminus to the C-terminus — remember this for translation.

The 20 amino acids

Amino acid	3-letter	1-letter	Side chain	In the diet
Alanine	Ala	A	Non-polar	Non-essential
Arginine	Arg	R	Basic (+)	Conditionally essential
Asparagine	Asn	N	Polar	Non-essential
Aspartic acid	Asp	D	Acidic (–)	Non-essential
Cysteine	Cys	C	Polar, forms S–S bridges	Conditionally essential
Glutamic acid	Glu	E	Acidic (–)	Non-essential
Glutamine	Gln	Q	Polar	Conditionally essential
Glycine	Gly	G	Non-polar (smallest)	Conditionally essential
Histidine	His	H	Basic (+)	**Essential**
Isoleucine	Ile	I	Non-polar	**Essential**
Leucine	Leu	L	Non-polar	**Essential**
Lysine	Lys	K	Basic (+)	**Essential**

Amino acid	3-letter	1-letter	Side chain	In the diet
Methionine	Met	M	Non-polar (start amino acid)	**Essential**
Phenylalanine	Phe	F	Non-polar, aromatic	**Essential**
Proline	Pro	P	Non-polar (ring)	Conditionally essential
Serine	Ser	S	Polar	Non-essential
Threonine	Thr	T	Polar	**Essential**
Tryptophan	Trp	W	Non-polar, aromatic	**Essential**
Tyrosine	Tyr	Y	Polar, aromatic	Conditionally essential
Valine	Val	V	Non-polar	**Essential**

Essential, non-essential and conditionally essential

- **Essential (9):** the body cannot make them, so they must come from food — histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. Memory trick: **PVT TIM HiLL** (Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Leucine, Lysine).
- **Non-essential:** the body can make them even if we do not eat them (for example alanine, aspartic acid, glutamic acid, asparagine, serine).
- **Conditionally essential (6):** normally made by the body, but during illness, stress or fast growth the body cannot make enough — arginine, cysteine, glutamine, glycine, proline and tyrosine.

SORTING THE LISTS
Conditionally essential amino acids are a sub-group of the "non-essential" ones. Essential (9) + the rest (11) = 20.

3. Nucleic acids: DNA and RNA

Nucleic acids carry the information that makes every living thing what it is. They control how proteins are made, so they decide the inherited traits of the organism. The two main types are DNA (deoxyribonucleic acid) and RNA (ribonucleic acid).

Building blocks

A nucleic acid is a long chain (a polynucleotide) of units called nucleotides. Each nucleotide has three parts:

- a five-carbon (pentose) **sugar** — deoxyribose in DNA, ribose in RNA;
- a **phosphate group**;
- a nitrogen-containing **base**.

A nucleoside is the sugar plus the base only — no phosphate. So: nucleoside + phosphate = nucleotide. In DNA the sugar is called 2′-deoxyribose because it lacks one hydroxyl (–OH) group that ribose has.

There are five bases. A (adenine) and G (guanine) are purines (two rings); C (cytosine), T (thymine) and U (uracil) are pyrimidines (one ring). DNA uses A, C, G, T. RNA uses A, C, G, U.

Nucleotides are joined into a chain by phosphodiester bonds: a phosphate bridges the 3′-hydroxyl of one sugar and the 5′-hydroxyl of the next. This gives the chain a direction — one end is the 5′ end and the other the 3′ end. New nucleotides are always added at the 3′ end, so nucleic acids are always built 5′ → 3′.

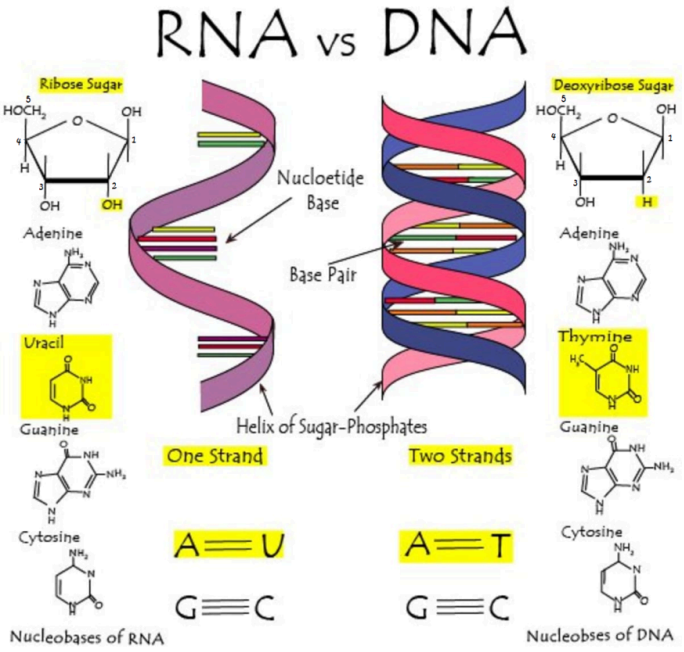


Figure 2. RNA compared with DNA: sugars, bases and number of strands.

	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Strands	Usually two (double helix)	Usually one
Where	Nucleus (and mitochondria, chloroplasts)	Made in the nucleus; works in the cytoplasm
Job	Long-term storage of genetic information	Copies, carries and decodes the information (mRNA, tRNA, rRNA)
Stability	Very stable	Short-lived, easily broken down

The double helix

DNA is two strands wound around each other. The sugar–phosphate backbones are on the outside and the bases point inward. The bases pair by hydrogen bonds: A pairs with T

(two hydrogen bonds) and G pairs with C (three hydrogen bonds). Each pair has nearly the same width, so the helix is uniform. The two strands run in opposite directions (antiparallel): one 5′ → 3′, the other 3′ → 5′.

This pairing rule is the secret of heredity: each strand is a template for the other, so DNA can be copied accurately. Its chemical stability makes DNA an ideal store of genetic information.

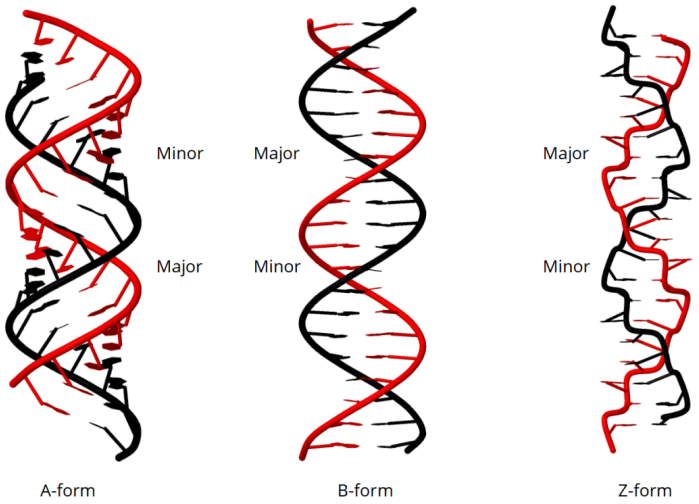


Figure 3. Three forms of DNA. B-form (centre) is the usual form inside cells; A-form is more compact; Z-form is a left-handed zig-zag. *Source: Dmitry M's training site*

A SHORT HISTORY (OPTIONAL)

1869 — Friedrich Miescher isolates "nuclein" from cell nuclei. 1944 — Avery, MacLeod and McCarty show DNA carries heredity. 1953 — Watson and Crick publish the double-helix model, using X-ray images made by Rosalind Franklin and Raymond Gosling (the famous Photo 51) and data from Maurice Wilkins. 1962 — Watson, Crick and Wilkins share the Nobel Prize. Franklin had died in 1958 and could not be included.

Good discussion point: whose work is remembered, and why?

4. The central dogma

The central dogma of molecular biology (Francis Crick, 1958) says that genetic information flows from DNA to RNA to protein, and that it cannot flow back from protein to nucleic acid. (Crick used "dogma" to mean "central idea".)

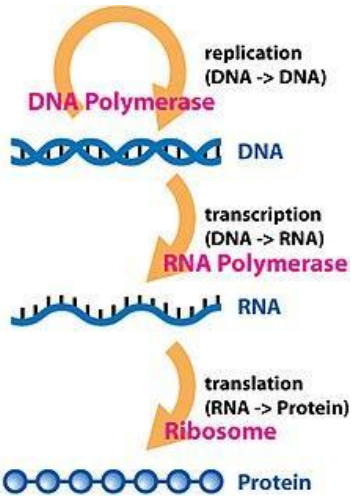


Figure 4. Replication (DNA → DNA), transcription (DNA → RNA) and translation (RNA → protein). *Source: Wikipedia*

Process	From → to	Main enzyme / machine
Replication	DNA → DNA	DNA polymerase
Transcription	DNA → RNA	RNA polymerase
Translation	RNA → protein	Ribosome

EXCEPTIONS THAT PROVE THE RULE
Some viruses (such as HIV) copy their RNA into DNA using reverse transcriptase, and many RNA viruses copy RNA into RNA. These are extra routes between nucleic acids — but information never goes from protein back into nucleic acid.

4.1 Replication — copying DNA

Before a cell divides it copies all of its DNA, so each new cell gets a complete set of chromosomes. In eukaryotes this happens in the nucleus during the S phase of the cell cycle. Replication is semi-conservative: each new DNA molecule has one old strand and one new strand (shown by Meselson and Stahl in 1958).

1. **Opening.** Replication starts at special sites called **origins of replication**. The enzyme **helicase** unwinds the double helix by breaking the hydrogen bonds, making a Y-shaped **replication fork**.
2. **Priming.** DNA polymerase cannot start from nothing. The enzyme **primase** first makes a short **RNA primer**.
3. **Copying. DNA polymerase** adds nucleotides to the primer, following the base-pairing rules. It can only add to a 3′ end, so it always builds **5′ → 3′**.
4. **Sealing. DNA ligase** joins the fragments and seals gaps in the sugar–phosphate backbone.

Leading and lagging strands

Because the two strands are antiparallel and the polymerase works only 5′ → 3′, the two new strands are made differently:

- The **leading strand** is copied continuously, in the same direction as the fork opens.
- The **lagging strand** is copied in short pieces called **Okazaki fragments**, each made in the direction *away* from the fork. Each fragment needs its own primer; ligase then joins them.

DNA polymerase also proofreads what it writes. Together with repair systems, this brings the error rate down to about one mistake in a billion letters.

Enzyme	Job in replication
Helicase	Unwinds the helix and breaks hydrogen bonds
Primase	Makes the short RNA primer
DNA polymerase	Adds nucleotides 5′ → 3′ and proofreads
Ligase	Joins Okazaki fragments, seals gaps

Base pairing in action

Take this parental DNA strand and see what each process makes from it:

```
5′-GAATTCCAAA-3′   parent strand (the template)
3′-CTTAAGGTTT-5′   new DNA strand made by replication
3′-CUUAAGGUUU-5′   RNA made by transcription (U instead of T)
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Read the RNA in the usual 5′ → 3′ direction and it is UUUGGAAUUC. Every strand follows the same pairing rules: A–T (or A–U in RNA) and G–C.

SPOT THE PATTERN
GAATTC reads the same on both strands when read 5′ → 3′ (a palindrome). It is also the cutting site of the enzyme EcoRI — you will meet it again on Day 3 when we cut and paste DNA.

4.2 Transcription — making an RNA copy

Transcription is the making of RNA from a DNA template by RNA polymerase. The enzyme unwinds a short stretch of DNA, reads the template strand 3′ → 5′, and builds an RNA chain 5′ → 3′. Unlike DNA polymerase it needs no primer. The RNA has the same sequence as the coding (non-template) strand, with U instead of T.

Many RNA copies can be made from one gene at the same time, and which genes are copied depends on the cell type and its environment. Transcription is less accurate than replication (roughly one error in 10,000–100,000 letters), but this matters little because RNA is temporary and many copies are made.

	Bacteria	Eukaryotes (e.g. humans)
Where	Cytoplasm	Nucleus
RNA polymerases	One	Three (I, II, III); polymerase II makes mRNA

	Bacteria	Eukaryotes (e.g. humans)
Finding the start	Sigma factor helps the polymerase bind the promoter (−10 and −35 regions)	Many transcription factors gather at the promoter; the TATA box lies about 25–30 bases upstream
DNA packaging	Not an issue	Chromatin-remodelling proteins open DNA wound around histones
Stopping	At a terminator sequence, often forming a hairpin loop	After a polyadenylation signal; the RNA is cut
After transcription	mRNA is ready; ribosomes can start translating while it is still being made	RNA is processed first (cap, tail, splicing), then exported

Processing of eukaryotic mRNA

- **5′ cap** — a modified guanine (7-methylguanosine) added to the front. It protects the RNA and helps the ribosome recognise it.
- **Poly-A tail** — a string of about 200 adenines added to the end. It also protects the RNA and helps its export.
- **Splicing** — the non-coding **introns** are cut out and the **exons** are joined. A large RNA–protein machine called the **spliceosome** does this.
- **Export** — only complete, spliced mRNA passes through the nuclear pore into the cytoplasm.

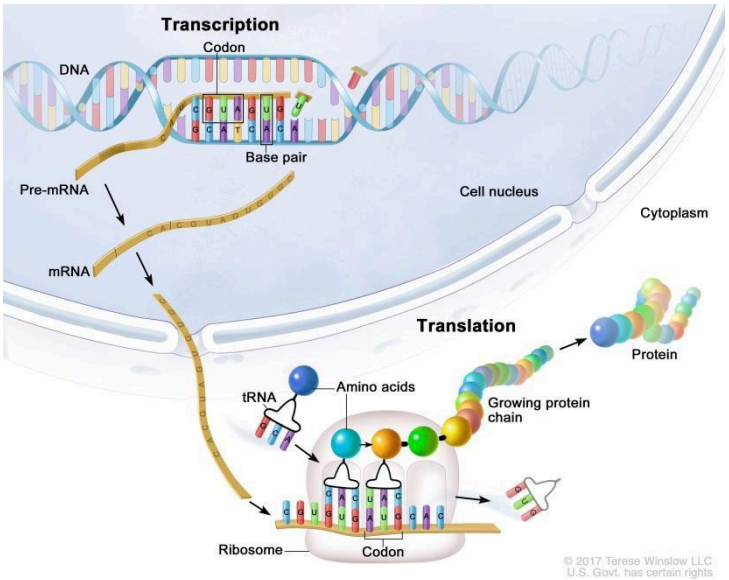


Figure 5. From gene to protein in a eukaryotic cell: transcription in the nucleus, processing of pre-mRNA, translation on the ribosome. *Source: NCI Dictionary of Genetics Terms*

4.3 Exons, introns and splicing

- **Exons** are the parts of a gene that are *kept* in the mature RNA. Most of an mRNA's coding sequence — and its start and stop codons — lie in exons. (Exons also include the untranslated regions at the two ends, called the 5′ UTR and 3′ UTR.)

- **Introns** are the "intervening" sequences that are *removed* from the pre-mRNA by splicing. They range from tens to thousands of bases. Introns occur in most organisms, including some viruses.
- Protein-coding exons make up only about **1–2 %** of the human genome.

Splicing is useful: the exons of one gene can be joined in different combinations (alternative splicing), so one gene can make several different proteins. This is one reason humans make far more proteins than they have genes.

Type of intron	How it is removed
In protein-coding genes	By the spliceosome
In tRNA genes	By protein enzymes
Group I self-splicing introns	The RNA cuts itself out, using a G nucleotide (GTP) as helper
Group II self-splicing introns	The RNA cuts itself out without GTP

WHY MUST SPLICING BE EXACT? A SENTENCE ANALOGY

Codons are read three letters at a time. Read this sentence in threes: THE CAT SAW THE DOG

Now delete one letter from the start: HEC ATS AWT HED OG — everything after the mistake becomes nonsense. This is a frameshift. If an intron is cut out a little too short or too long, the same thing happens to the protein.

5. Translation — making the protein

Translation is the decoding of the message in mRNA into a chain of amino acids. It happens in the cytoplasm on ribosomes, and like transcription it has three stages: initiation, elongation and termination.

Component	Role
mRNA	The message: a sequence of codons
Ribosome	The decoding machine made of rRNA and protein. Eukaryotes: 80S (40S + 60S subunits). Bacteria: 70S (30S + 50S).
tRNA	The adaptor. One end carries an amino acid; the other has an anticodon that pairs with a codon on the mRNA
Aminoacyl-tRNA synthetases	Enzymes that attach the correct amino acid to each tRNA ("charging")
GTP / ATP	Energy

The genetic code

- A **codon** is three bases. There are 4 × 4 × 4 = **64** codons: 61 code for amino acids and 3 are **stop** codons (UAA, UAG, UGA).
- **AUG** codes for methionine and is the usual **start** codon.
- The code is **redundant**: most amino acids have more than one codon (for example GCU, GCC, GCA and GCG all mean alanine).
- The code is read in order, without gaps or overlaps, and is almost the same in all living things.

1st ↓ 2nd →	**U**	**C**	**A**	**G**
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly

HOW TO READ THE TABLE

For the codon GAA: find first base G (row), second base A (column), then the third base A inside the box → Glu (glutamic acid). AUG (Met) is the start codon; UAA, UAG and UGA are the stop codons.

The three stages

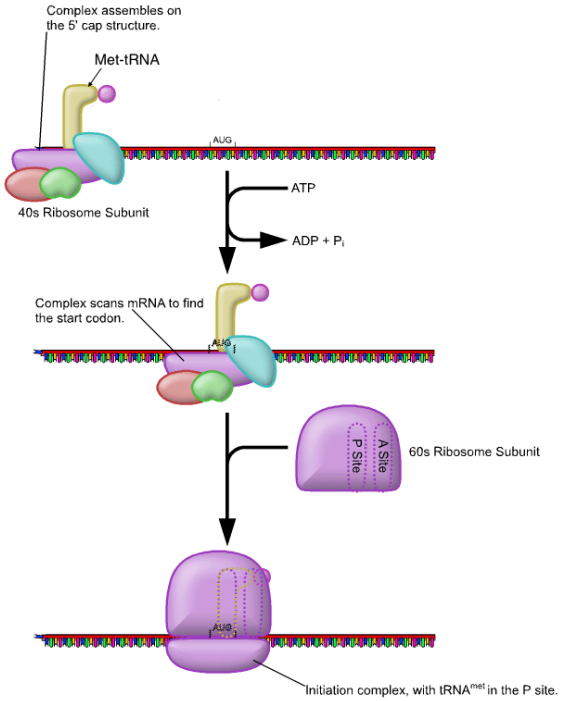


Figure 6. Initiation of translation in eukaryotes: the small ribosomal subunit binds the 5′ cap, scans to the start codon AUG, and the large subunit joins. *Source: TeachMePhysiology*

Initiation

The small ribosomal subunit (40S) binds the 5′ cap of the mRNA, carrying the first tRNA (Met-tRNA). It scans along until it finds the first AUG. The large subunit (60S) then joins and the complete ribosome is ready. (In bacteria the small subunit finds the start by a sequence called the Shine–Dalgarno sequence.)

Elongation

The ribosome has three tRNA sites: A (arrival), P (peptide) and E (exit).

1. The tRNA carrying the first amino acid (Met) sits in the **P site**.
2. A charged tRNA whose anticodon matches the next codon enters the **A site** (this uses energy from GTP).
3. The ribosome forms a **peptide bond**: the growing chain is moved from the tRNA in the P site onto the new amino acid held in the A site.
4. The ribosome **translocates** (moves) exactly one codon along the mRNA. The now-empty tRNA moves to the E site and leaves; the tRNA holding the chain is now in the P site; the A site is free for the next tRNA.
5. Steps 2–4 repeat for every codon. The chain grows from the **N-terminus** (Met) to the **C-terminus**.

Termination

When a stop codon (UAA, UAG or UGA) reaches the A site, no tRNA matches it. Instead, proteins called release factors bind, and the finished polypeptide is released. The ribosome subunits separate and can be used again. Many ribosomes can read the same mRNA at once, like cars following each other on a road.

WORKED EXAMPLE: FROM GENE TO PEPTIDE

Coding strand of DNA (5′ → 3′): ATG GCT TGT GAA AAG CTT TAA

mRNA (5′ → 3′): AUG GCU UGU GAA AAG CUU UAA

Amino acids: Met – Ala – Cys – Glu – Lys – Leu – Stop

Peptide in one-letter code: M A C E K L — a six-amino-acid peptide.

The template strand, written 3′ → 5′, would be TAC CGA ACA CTT TTC GAA ATT.

After translation: finishing the protein

A new polypeptide is rarely ready to work. It must fold, and is often modified. These changes are called post-translational modifications.

Change	What happens	Example
Folding	Helper proteins (chaperones) guide the chain into its 3D shape	Almost every protein
Cleavage	Part of the chain is cut off to activate it	Insulin is made as a longer chain and trimmed
Phosphorylation	A phosphate group is added; works as an on/off switch	Many signalling proteins
Glycosylation	Sugar chains are attached	Many proteins on the cell surface
Disulfide bridges	S–S bonds form between cysteines and lock the shape	Antibodies
Tagging for removal	Ubiquitin tags mark a protein for destruction	Damaged or no-longer-needed proteins

Check your understanding

1. Which part of an amino acid is different in each of the 20 amino acids?
2. Why is the nucleoside adenosine not a nucleotide?
3. A DNA template strand reads 3′-TACGGA-5′. Write the mRNA and the amino acids it codes for.
4. Which enzyme makes the RNA primer during replication? Why is a primer needed?
5. Why does a missing intron-removal step produce a faulty protein?
6. Name the three tRNA sites on the ribosome and say what each does.

Answers are in the Quizzes and Answer Keys document.